

Lesson 06 Demo 01

Configuring Virtual Machine Availability Set

Objective: To set up an availability set for a virtual machine to ensure high availability and reliability

Tools Required: Azure account with admin access

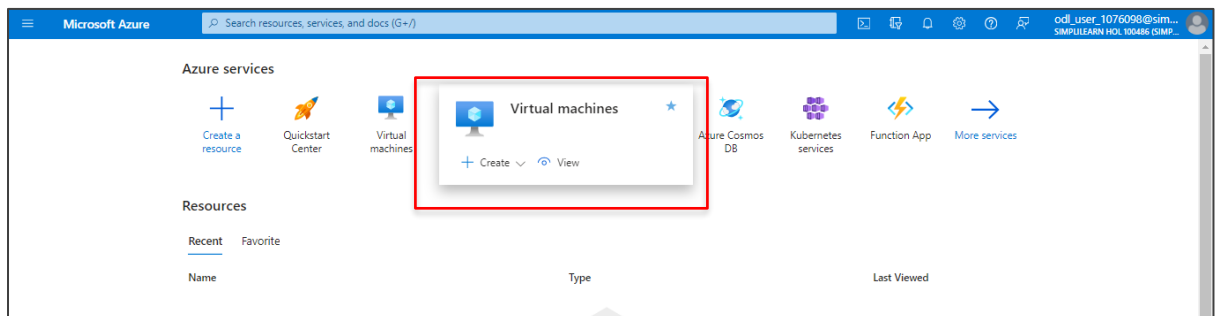
Prerequisites: None

Steps to be followed:

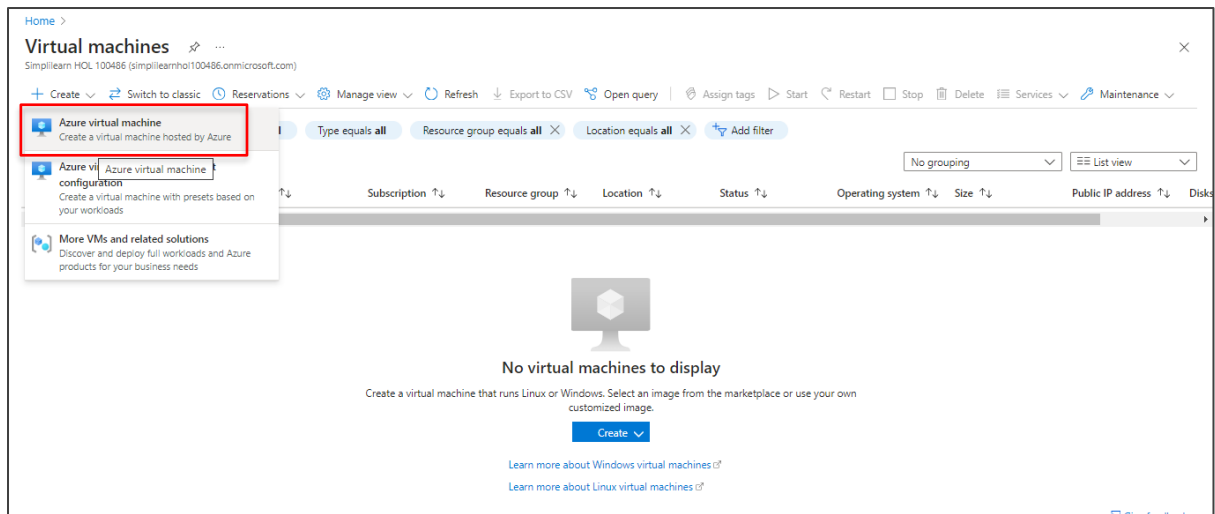
1. Configure the availability set while creating a VM

Step 1: Configure the availability set while creating a VM

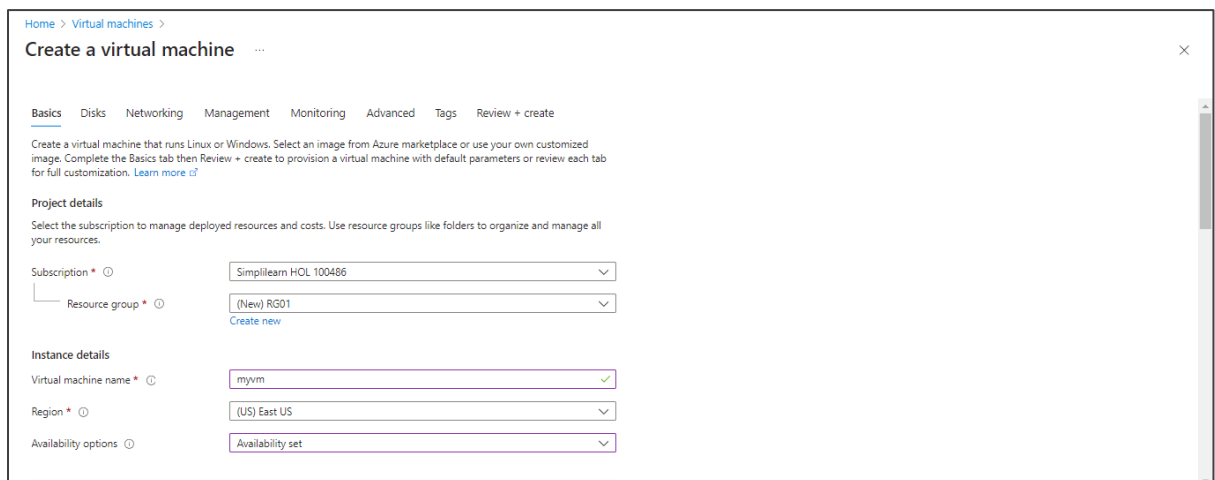
- 1.1 Log in to the **Azure portal** and search for **Virtual machines**



1.2 Click + Create and select Azure virtual machine



1.3 In the Basics tab, click Create new to create a new Resource group. Now, enter the other details as shown in the screenshot below:



1.4 In the **Create a virtual machine** page, click **Create new** under the **Availability set**. Now, under the **Create availability set** page, enter **Name** as **MyAvlSet** and click **OK**.

The screenshot shows two side-by-side panels from the Azure portal. The left panel, titled 'Create a virtual machine', has the 'Availability set' dropdown set to 'No existing availability sets in current resource group and location.' with a 'Create new' link below it. Other settings include 'Security type' as 'Trusted launch virtual machines', 'Image' as 'Ubuntu Server 20.04 LTS - x64 Gen2', 'VM architecture' as 'x64', and 'Size' as 'Standard_D2s_v3 - 2 vcpus, 8 GiB memory (Price unavailable)'. The right panel, titled 'Create availability set', shows the 'Name' field filled with 'MyAvlSet', 'Fault domains' set to 2, and 'Update domains' set to 5. The 'Use managed disks' option is set to 'Yes (Aligned)'. Both panels have 'OK' buttons at the bottom.

1.5 Choose the **Security type** and **Image**, and leave the other options as default, enter **Administrator account's username**

The screenshot shows the 'Create a virtual machine' panel with updated settings. The 'Availability set' dropdown now shows '(new) MyAvlSet'. The 'Security type' is set to 'Standard'. The 'Image' is set to 'Windows Server 2022 Datacenter: Azure Edition Hotpatch - x64 Gen2'. The 'VM architecture' is still 'x64'. The 'Size' remains 'Standard_D2s_v3 - 2 vcpus, 8 GiB memory (Price unavailable)'. Under the 'Administrator account' section, the 'Username' field is now filled with 'simpli'.

1.6 Enter **Password** and **Confirm password**, choose **HTTP (80)**, **SSH (22)**, and **RDP(3389)**, and click on **Next: Disks >** under **Select inbound ports**

Home > Virtual machines >

Create a virtual machine

Password * ✓

Confirm password * ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ☐ None ☒ Allow selected ports

Select inbound ports *

ⓘ All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

Licensing

Save up to 49% with a license you already own using Azure Hybrid Benefit. [Learn more](#)

Would you like to use an existing Windows Server license? ☐

[Review + create](#) [< Previous](#) [Next: Disks >](#) [Give feedback](#)

1.7 Select **Standard HDD (locally-redundant storage)** under **OS disk type**, and click **Review + create**

Home > Virtual machines >

Create a virtual machine

Basics **Disks** Networking Management Monitoring Advanced Tags Review + create

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host ☐

ⓘ Encryption at host is not registered for the selected subscription. [Learn more about enabling this feature](#)

OS disk

OS disk type *

The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM ☒

Key management

[Review + create](#) [< Previous](#) [Next: Networking >](#) [Give feedback](#)

1.8 Click Create

Home > Virtual machines >

Create a virtual machine

Validation passed

Basics Disks Networking Management Monitoring Advanced Tags **Review + create**

Cost given below is an estimate and not the final price. Please use [Pricing calculator](#) for all your pricing needs.

Price

1 X Standard D2s v3
by Microsoft
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Pricing not available for this offering

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; and (b) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Create < Previous Next > [Download a template for automation](#) [Give feedback](#)

The final output appears as shown below:

Home >

CreateVm-MicrosoftWindowsServer.WindowsServer-202-20230921123002 | Overview

Deployment

Search < Delete Cancel Redeploy Download Refresh

Your deployment is complete

Deployment name: CreateVm-MicrosoftWindowsServer.WindowsServer-202-20230921123002 Start time: 9/21/2023, 12:33:07 PM
Subscription: Simplilearn HOL 100486 Correlation ID: 6a0e41fc-f304-4152-b36e-1bb623fe06fc
Resource group: RG01

Deployment details

Next steps

- Setup auto-shutdown Recommended
- Monitor VM health, performance and network dependencies Recommended
- Run a script inside the virtual machine Recommended

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Home >

Virtual machines

Simplilearn HOL 100486 (simplilearnh100486@microsoft.com)

+ Create Switch to classic Reservations Manage view Refresh Export to CSV Open query Assign tags Start Restart Stop Delete Services Maintenance

Filter for any field... Subscription equals all Type equals all Resource group equals all Location equals all Add filter

Showing 1 to 1 of 1 records.

Name	Type	Subscription	Resource group	Location	Status	Operating system	Size	Public IP address	Disks
myvm	Virtual machine	Simplilearn HOL 100486	RG01	East US	Running	Windows	Standard_D2s_v3	172.190.137.67	1

By following the above steps, you have successfully set up an availability set for a virtual machine to ensure high availability and reliability.